Research on Single-Frequency PPP-B2b Time Transfer

Zaimin He, Lan Li, Runzhi Zhang, Juan Hou, Gongwei Xiao, Wei Guang, Jihai Zhang, and Xiangyi He

Time transfer based on global navigation satellite system (GNSS) using precise point positioning (PPP) is a critical technique in universal time coordinated (UTC) calculation. PPP relies on precise satellite orbit and satellite clock offset products obtained through network solutions. Since August 2020, the Beidou global navigation satellite system (BDS-3) has provided GNSS users with the PPP product using PPP-B2b signal through three geostationary earth orbit (GEO) satellites instead of a network in the Asia-Pacific area. Network interruption will cause PPP reconvergence and terminate the time transfer, which hinders the application of PPP time transfer. The real-time PPP-B2b product broadcast by GEO satellites can be considered to solve the problem of interruption caused by network fluctuation in the field of real-time PPP. To promote the application of real-time PPP time comparison in UTC calculation and develop PPP-B2b application in time transfer, single-frequency (SF) PPP-B2b time transfer is investigated. Meanwhile, the treatment of ionospheric delay of SF PPP-B2b is crucial, and a GRAPHIC SF PPP-B2b model is proposed and validated in order to keep PPP-B2b network-independent. Data from two international atomic time (TAI) time-keeping laboratories and two international GNSS service (IGS) stations were used to analyze the accuracy in SF time comparison applications using the PPP-B2b product from common clock difference (CCD) and long-baseline time links. Results show that the SF PPP-B2b CCD sequences can be concentrated within 0.2 ns. The long-baseline time difference between SF time transfer using PPP-B2b product and dual-frequency (DF) time transfer using the final product fluctuated within 2 ns. In addition, the standard deviation (STD) values of PPP-B2b time comparison are mainly distributed at 0.5 ns. Meanwhile, only BDS and BDS + GPS can achieve similar SF PPP-B2b time transfer accuracy.

Background

The high-precision time scale can be generated and maintained by the time-frequency system, and the high-precision time transfer is an indispensable part of the time-frequency system. Presently, GNSS time transfer mainly includes common-view (CV), all-in-view (AV), and GNSS carrier phase. Compared with CV and AV based on low-precision pseudorange observations, GNSS carrier phase time transfer has higher time transfer accuracy. It is realized by means of PPP [1] which uses a single GNSS receiver to obtain precise absolute coordinates. The GeoForschungZentrum (GFZ) analysis center product GBM, whose accuracy can reach 3 cm, can be obtained with a two-day delay. The STD value of post-process PPP time transfer using the GBM product can reach 0.3 ns [2]. Since 2013, the IGS has provided users with the precise satellite orbit and satellite clock offset products required by PPP for free via the networked transport of Radio Technical Commission for Maritime Services (RTCM) via internet protocol (NTRIP). Ge *et al.* analyzed real-time PPP time transfer [3]. Their results showed that real-time PPP time transfer could achieve a 0.5 ns level. However, the real-time PPP via the NTRIP protocol can be performed only in smooth network environments. And network interruption causes instability in PPP time transfer relying on IGS products. Since August 2020, precise products called the PPP-B2b product have been available through three GEO satellites instead of NTRIP protocol [4]. Presently, the research on the PPP-B2b product focuses on product quality, positioning, and DF time transfer [5],[6]. The DF time comparison using the PPP-B2b product can reach an accuracy of sub-nanosecond. The price of the SF receiver has attracted more and more researchers to study SF PPP time transfer. However, the PPP-B2b time transfer only verifies the feasibility of dual-frequency receiver, while the accuracy of time transfer of single-frequency receiver is still blank. The advantages of SF PPP-B2b time transfer receiver in cost and stability are beginning to be realized. The treatment of PPP ionospheric delay is very tricky. For dual-frequency, the effect of ionospheric delay can be eliminated by the frequency combination, while single-frequency cannot directly eliminate the effect of ionospheric delay. Also, further PPP solution of the PPP-B2b product is independent of a network, so that the external input of ionosphere is isolated. An analysis of performance of SF PPP-B2b time transfer is required.

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In this study, a new single-frequency PPP-B2b time transfer was investigated. Two TAI timekeeping laboratories, including the National Time Service Center of the Chinese Academy of Sciences (NTSC) and the Telecommunication Laboratories (TL), external atomic clock IGS station (USUD) and an IGS station (CUSV) with an internal clock were selected to validate the accuracy of the PPP-B2b product in SF comparison applications from zero-baseline CCD and long-baseline time links.

Method

Currently, the PPP-B2b product provides satellite orbit and satellite clock offset correction of navigation ephemeris for BDS-3 and GPS users, respectively.

Precise Satellite Orbit

The rough satellite position which is calculated through the broadcast ephemeris is corrected using the PPP-B2b product and (1) to obtain the precise satellite position:

$$\begin{bmatrix} X_{orbit} \\ Y_{orbit} \\ Z_{orbit} \end{bmatrix} = \begin{bmatrix} X_{brdc} \\ Y_{brdc} \\ Z_{brdc} \end{bmatrix} - [e_{radial}\ e_{along}\ e_{cross}] \cdot \begin{bmatrix} \delta O_R \\ \delta O_A \\ \delta O_C \end{bmatrix} \tag{1}$$

In (1), $\left[X_{orbit}\ Y_{orbit}\ Z_{orbit}\right]^T$ represents the precise satellite position vector; $\left[X_{brdc}\ Y_{brdc}\ Z_{brdc}\right]^T$ indicates the real-time satellite position calculated from the ephemeris; $[e_{radial}\ \ e_{along}\ \ e_{cross}]$ corresponds to the unit vectors which can be calculated as (2) in radial, along-track, and cross-track directions, respectively; and $\left[\delta O_R\ \delta O_A\ \delta O_C\right]^T$ represents the orbit correction vector from the PPP-B2b product:

$$\begin{cases} e_{radial} = \dfrac{r}{|r|} \\ e_{cross} = \dfrac{r \times v}{|r \times v|} \\ e_{along} = e_{cross} \times e_{radial} \end{cases} \tag{2}$$

In (2), r and v are the satellite position and velocity vectors of ephemeris.

Precise Satellite Clock

The precise satellite clock offset is obtained by correcting the satellite clock offset of ephemeris through the PPP-B2b product and (3):

$$dt^s_{prec} = dt^s_{brdc} - \frac{C_0}{c} \tag{3}$$

In (3), dt^s_{prec} indicates the precise clock offset; dt^s_{brdc} indicates the satellite clock offset calculated from ephemeris; C_0 indicates the clock correction provided by the PPP-B2b product in meters; and c indicates the speed of light.

GRAPHIC SF PPP-B2b Model

The biggest challenge for SF PPP is the treatment of ionospheric delay. Averaging SF pseudorange observations and carrier phase observations to remove first-order ionospheric delays is called GRAPHIC [7]. Meanwhile, combining with broadcast ionospheric products such as the Beidou global broadcast ionospheric delay correction model (BDGIM) [8] to replace IGS ionospheric products [9], constraining the ionospheric delay in the pseudorange observation equation with the broadcast ionosphere model, a new SF PPP-B2b time transfer model was investigated. The ionosphere-free SF PPP-B2b model can be written as follows:

$$\begin{cases} p^s_{r\_iono,j} = \mu^s_r \cdot x + c \cdot d\tilde{t}_r + I^s_{r,j} + T^s_r + \varepsilon^s_{r\_iono,j} \\ l^s_{IF,j} = \dfrac{p^s_{r,j} + l^s_{r,j}}{2} = \mu^s_r \cdot x + c \cdot d\tilde{t}_r + T^s_r + \lambda \tilde{N}^s_{r,j} + \xi^s_{IF,j} \\ Ion^s = I^s_{r,j} + \zeta_{ion} \end{cases} \tag{4}$$

In (4), $p^s_{r\_iono,j}$ refers to pseudorange observation corrected using the differential code bias PPP-B2b product; μ^s_r denotes the receiver-to-satellite unit vector; c is the speed of light in vacuum, $\mathbf{x}$ denotes the vector of receiver position increments; $d\tilde{t}_r$ refers to the receiver clock offset in seconds; $I^s_{r,j}$ refers to the ionospheric delay in meters; T^s_r is the tropospheric delay in meters; $\varepsilon^s_{r\_iono,j}$ refers to the pseudorange observation noise; $p^s_{r,j}$ and $l^s_{r,j}$ refer to the raw observations minus the computed values of the carrier phase and pseudorange; $\lambda \tilde{N}^s_{r,j}$ represents the ambiguity in meters; $\xi^s_{IF,j}$ represents ionospheric-free SF observation noise; Ion^s denotes ionospheric delay calculated from broadcast ephemeris; and ζ_{ion} denotes broadcast ephemeris ionospheric measurement noise. Thus, the SF PPP-B2b time transfer estimated parameter vector $\bar{X}$ is:

$$\bar{X} = \begin{bmatrix} x & d\tilde{t}_r & T^s_r & I^s_{r,j} & \lambda \tilde{N}^s_{r,j} \end{bmatrix}. \tag{5}$$

Process and Strategies

SF PPP using the PPP-B2b product is performed as shown in Fig. 1. Table 1 lists SF PPP-B2b processing strategies. The PPP processing process is divided into data preparation, data preprocessing, error correction, Kalman filtering and estimation. Navigation ephemeris and the PPP-B2b product are used to recover precise satellite orbit and clock offset, and raw observations are also required for PPP calculation. Data preprocessing consists of gross error, clock slip detection reparation and cycle slip detection to provide pure raw data for subsequent high-precision algorithms. Errors are processed by models, empirical formulas and parameters estimated. The parameters estimated after Kalman filtering are verified by residual and the receiver clock offset can be obtained.

PPP Time Transfer

The receiver clock offset can be considered as time difference between the receiver local time t_{loc} and the GNSS reference time t_{ref}. The GNSS reference time of the two stations through time transfer is eliminated by the difference, which is written as (6):

$$\Delta t = d\tilde{t}_{r,1} - d\tilde{t}_{r,2} = (t_{loc,1} - t_{ref}) - (t_{loc,2} - t_{ref}) = t_{loc,1} - t_{loc,2} \tag{6}$$

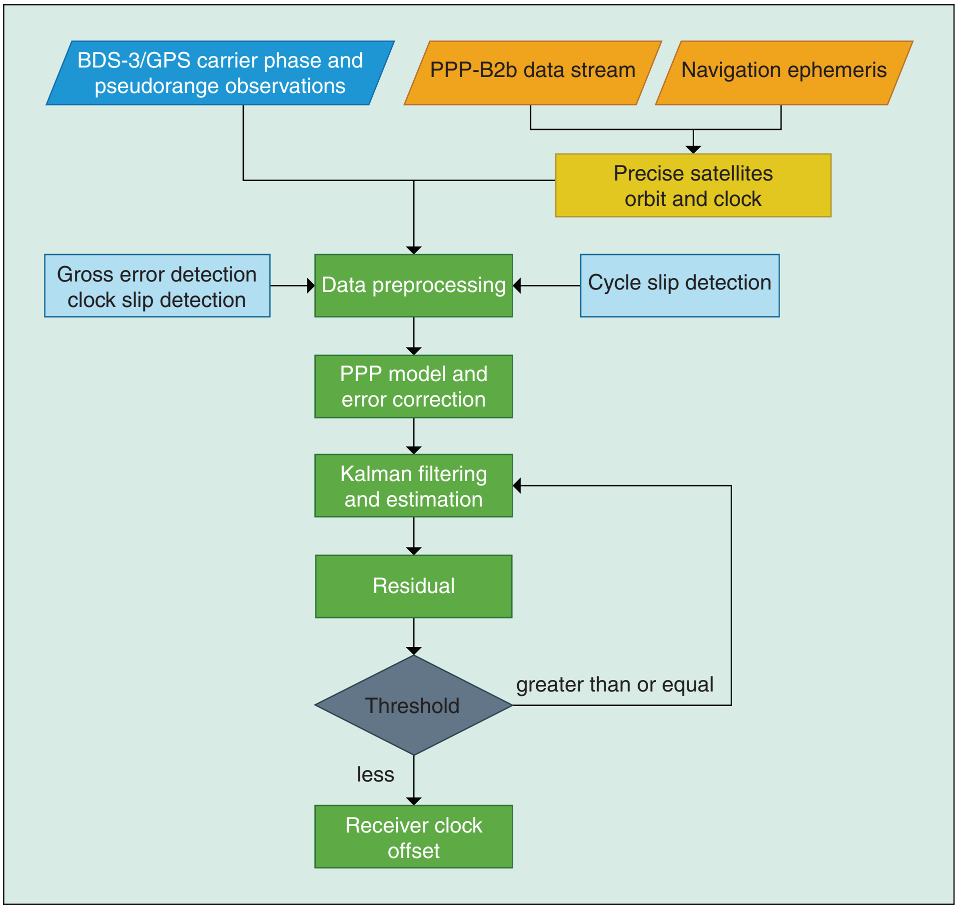

Fig. 1. Procedure for data processing.

Results and Discussion

The CCD [12],[13] experiment which is often selected to analyze the noise level of time transfer is implemented with observations from two GNSS receivers, named NTTS and NT07, with common antenna and external clock. The observations participating zero-baseline CCD were used for 11 d from day of year (DoY) 86 to 96 in 2022. The PPP-B2b product binary files were saved by the SINO K803 kit, which was equipped with PPP-B2b raw binary receiving module. Moreover, three time comparison long-baseline links are formed. Fig. 2 shows their geographic location. Table 2 lists detailed information of receivers. PPP time comparison using the GBM product is performed as a reference.

Table 1 – SF PPP-B2b processing strategies	
Item	**Strategies**
System	BDS, GPS
Estimator	Kalman filter
Cutoff elevation	10°
Raw observations	GPS: L1C/A; BDS: B3I
Sample interval	30 s
Satellite hardware delay correction	PPP-B2b differential code bias product
Tropospheric delay	Dry delay: corrected with Saastamoinen model [10] Wet delay: estimated as a random walk noise Mapping function: GMF [11]
Receiver antenna phase center	Corrected by the “igs*.atx” file
Receiver clock offset	Estimated as white noise
Phase ambiguity	Estimated as float

PPP-B2b CCD

Fig. 3 and Fig. 4 illustrate the receiver clock offset using GBM and PPP-B2b products, respectively. The receiver clock offset for time comparison using PPP is incorporated into the product reference time. Compared with GBM time comparison, the PPP-B2b product reference time contained in the receiver clock offset is continuous. As shown in Fig. 4, the receiver clock offset acquired by only GPS time transfer using the PPP-B2b product is intermittent. The GPS PPP-B2b product does not seem to maintain the satellite clock reference. Compared to the IGS product obtained from the global observation network, the

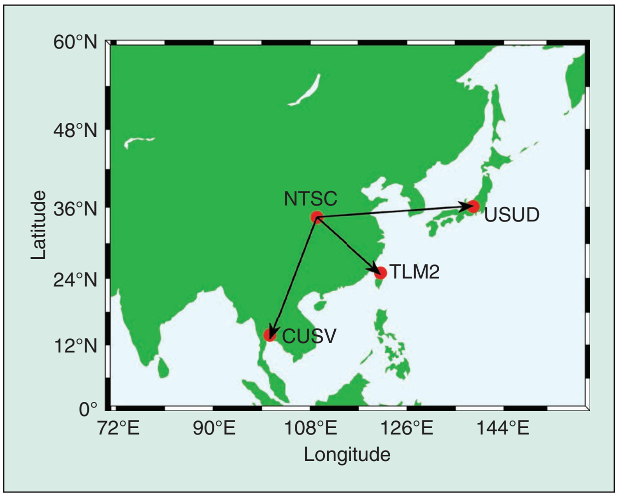

Fig. 2. The geographic distribution of time comparison links stations.

Table 2 – Detailed information of time links receivers

Station/ID	Organization	Receiver	Clock
NTTS	NTSC	SEPT POLARX5	External
NT07	NTSC	SEPT POLARX5	External
TLM2	TL	SEPT POLARX5	External
USUD	IGS	SEPT POLARX5	External H-Maser
CUSV	IGS	JAVAD TRE_3 DELTA	Internal

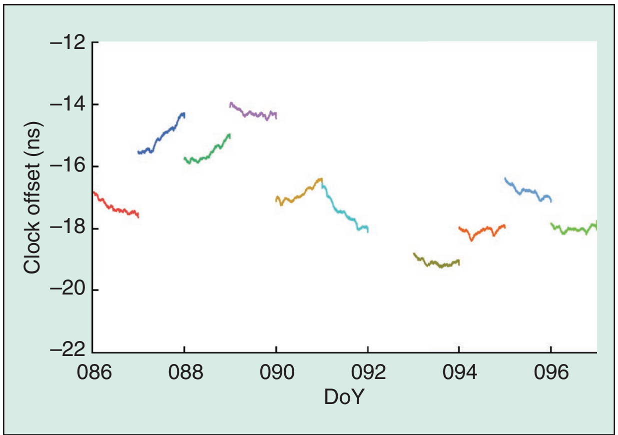

Fig. 3. NTTS clock offset using the GBM product for 11 days from DoY 86 to 96 in 2022.

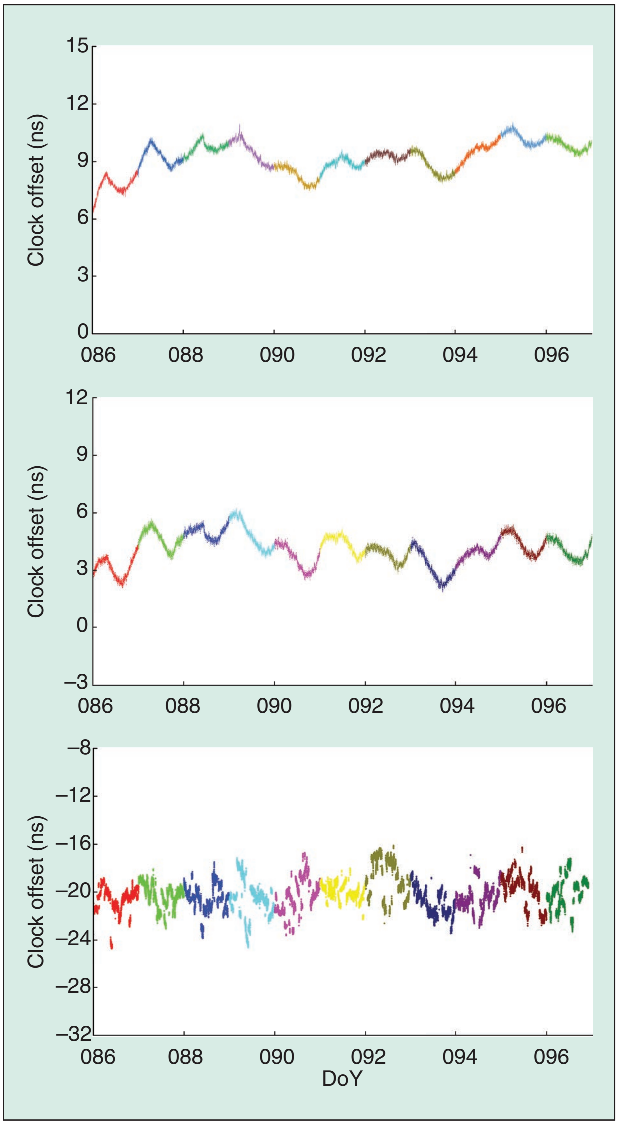

Fig. 4. PPP-B2b NTTS clock offset (GPS + BDS-3, BDS-3, GPS from top to bottom) for 11 days from DoY 86 to 96 in 2022.

PPP-B2b tracking stations are only in the territory of China. As a result, it is apparent that the time comparison of BDS-3 and BDS-3 + GPS PPP-B2b is better than that of a single GPS system. The content of this study is mainly based on BDS-3 and BDS-3 + GPS.

The zero-baseline CCD result is displayed in Fig. 5. The CCD result sequence of SF PPP-B2b fluctuates within 0.2 ns. The STD value of the CCD result residuals was calculated, and results showed that the addition of GPS system failed to improve the accuracy of SF PPP-B2b time transfer.

Long-baseline PPP-B2b Time Comparison

Taking the DF GBM time comparison as a reference, the differences among the long-baseline time comparison results are shown in Fig. 6 and Fig. 7. Meanwhile, Table 3 lists the STD values of the residuals. Combining Fig. 6, Fig. 7 and Table 3, two conclusions have been drawn. First, taking the GBM time comparison as a reference, the PPP-B2b time comparison residual for all three links fluctuated within 2 ns. Furthermore, the receiver clock offset introduces the hardware delay which led to a constant deviation in different time links. The STD values of the time comparison residual of SF PPP-B2b were concentrated at 0.5 ns. Second, regardless of the three-time comparison long-baseline links of PPP-B2b, it could be observed that the addition of GPS system did not improve the accuracy of time comparison.

Conclusions

The emergence of the PPP-B2b product promotes the application of real-time PPP in positioning in a networkless occasion. Meanwhile, the PPP-B2b product broadcast by three GEO satellites solves the problem of interruption caused by network fluctuation in the field of real-time PPP time comparison. The treatment of ionospheric delay of SF PPP-B2b is crucial, and a GRAPHIC SF PPP-B2b model is proposed and validated to keep PPP-B2b network-independent. Considering that the PPP-B2b product serves GPS and BDS users in the Asia-Pacific area, two TAI laboratories (NTSC, TL), an external atomic clock station (USUD) and an IGS

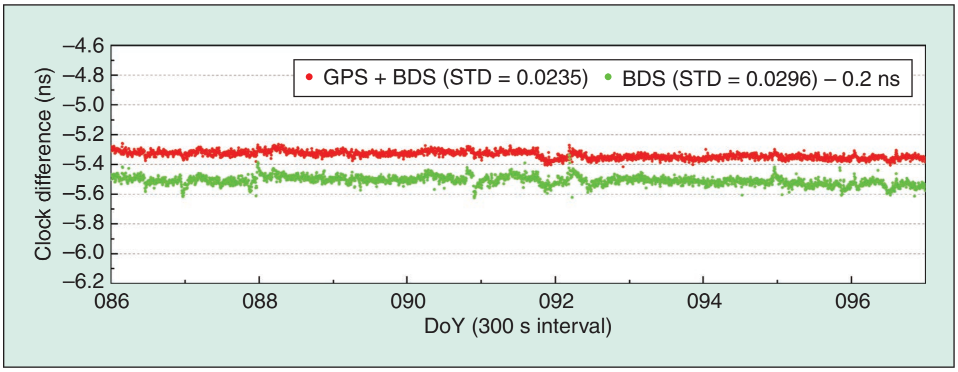

Fig. 5. PPP-B2b zero-baseline CCD for 11 days from DoY 86 to 96 in 2022.

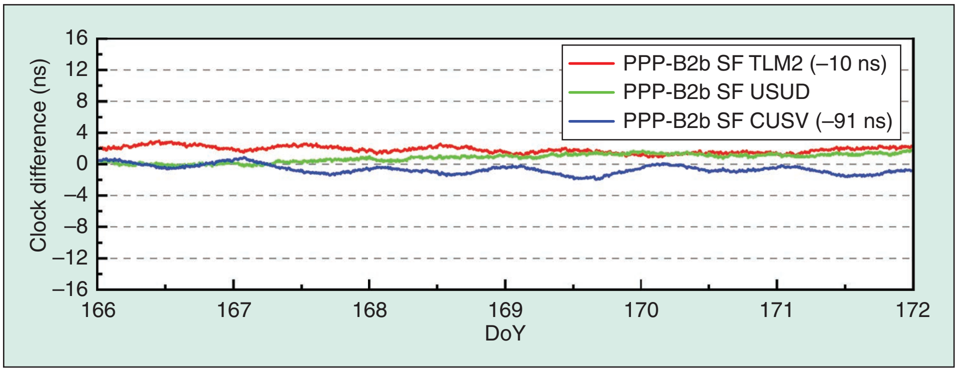

Fig. 6. BDS + GPS PPP-B2b time comparison residual.

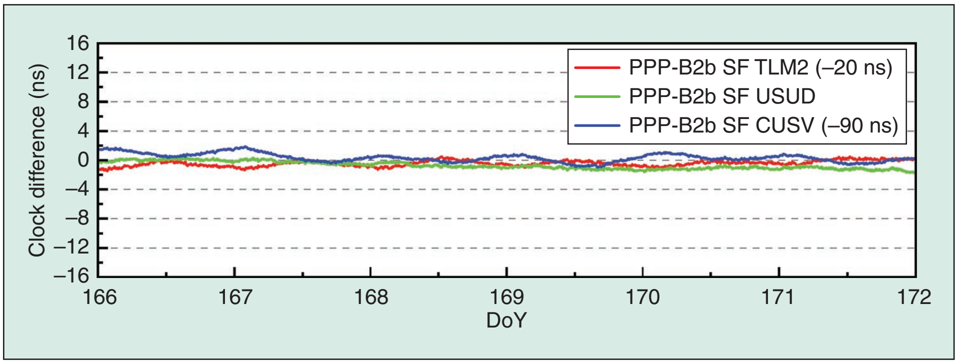

Fig. 7. BDS PPP-B2b time comparison residual.

Table 3 – STD values of PPP-B2b time comparison residual (ns)

Time link	Model	
	GPS + BDS	BDS
NTSC-TLM2	0.4287	0.3995
NTSC-USUD	0.5028	0.5112
NTSC-CUSV	0.5328	0.5425

station (CUSV) with an internal clock which were equipped with BDS-3 and GPS receivers were selected to validate the SF PPP-B2b time comparison from NTSC zero-baseline CCD and long-baseline time links. The GBM product was used as a reference to measure the accuracy of the time comparison using the PPP-B2b product. Compared with GBM time comparison, the reference time contained in the receiver clock offset using the PPP-B2b product is continuous. The receiver clock offset through GPS time transfer using the PPP-B2b product is intermittent. Only GPS SF PPP-B2b time comparison is not recommended, which is similar to DF PPP-B2b time transfer. The CCD result sequence of SF PPP-B2b fluctuates within 0.2 ns. Taking the GBM product as a reference, the STD values of SF PPP-B2b time comparison using GRAPHIC SF PPP-B2b model can be 0.5 ns. Meanwhile, the addition of GPS system does not significantly improve the PPP-B2b time transfer accuracy.

Acknowledgment

The authors acknowledge TL, IGS and GFZ for providing observations and precise satellite orbit and clock products. This research was funded by the State Key Laboratory of Geodesy and Earth's Dynamics (Grant No. SKLGED2022-3-3).

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Zaimin He (hezaimin@xupt.edu.cn) is a Professor at the School of Communications and Information Engineering and School of Artificial Intelligence, Xi'an University of Posts and Telecommunications, Xi'an, Shaanxi, China. He received the M.S. and Ph.D. degrees from the University of Chinese Academy of Sciences in 2008 and 2012, respectively. His main research fields are in radio positioning, timing, and communication.

Lan Li (lilan@stu.xupt.edu.cn) is a graduate student at the School of Communications and Information Engineering and School of Artificial Intelligence, Xi'an University of Posts and Telecommunications, Xi'an, Shaanxi, China. Her main research fields are high precision positioning and time transfer.

Runzhi Zhang (zhangrunzhi@ntsc.ac.cn) is currently working toward an M.S. degree in communication and information system with National Time Service Center, Chinese Academy of Sciences, Xi'an, China. His main research fields are time transfer and high precision positioning. He received a B.E. degree in electronic information engineering from Yanshan University, Qinhuangdao, China in 2020.

Juan Hou (hhoujuan@126.com) is an Engineer at Xi'an University of Posts and Telecommunications, Xi'an, Shaanxi, China. She received the M.S. degree from the University of Chinese Academy of Sciences in 2012. Her main research fields are in timekeeping techniques and the performance of atomic clocks.

Gongwei Xiao (xiaogongwei@xupt.edu.cn) is a Lecturer with the Xi'an University of Posts and Telecommunications, Xi'an, Shaanxi, China. His current research mainly focuses on multi-GNSS PPP, real-time precise orbit determination for LEO satellites and tropospheric delay. He received the M.S. degree from the College of Geodesy and Geomatics, Shandong University of Science and Technology in 2018 and received the Ph.D. degree from State Key Laboratory of Geodesy and Earth Dynamics, Innovation Academy for Precision Measurement Science and Technology, Chinese Academy of Sciences in 2021.

Wei Guang (weiguang@foxmail.com) is a Lecturer at the School of Communications and Information Engineering and School of Artificial Intelligence, Xi'an University of Posts and Telecommunications, Xi'an, Shaanxi, China. He received the M.S. and Ph.D. degrees from University of Chinese Academy of Sciences in 2012 and 2019, respectively. His main research field is high performance time transfer.

Jihai Zhang (zhangntsc@126.com) is a Research Associate at the National Time Service Center of Chinese Academy of Sciences, Xi'an, Shaanxi, China. He received the M.S. degree in 2014 from the University of Chinese Academy of Sciences. Now he is a Ph.D. student at the University of Chinese Academy of Sciences. His main research fields are high precision time transfer and GNSS time difference monitoring.

Xiangyi He (hexiangyiii@163.com) is a graduate student at the School of Communications and Information Engineering and School of Artificial Intelligence, Xi'an University of Posts and Telecommunications, Xi'an, Shaanxi, China. Her main research fields are in GNSS, troposphere.